Billy Lukito Danuharja

Computer Engineering Graduate | Robotics & Embedded Systems

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SUMMARY

Robotics and Embedded Systems Engineer with hands-on experience developing autonomous systems across ground robots and unmanned aerial vehicles. Skilled in integrating electronic hardware, embedded firmware, sensors, control algorithms, and robotics software into reliable working systems. Project experience includes adaptive cruise control for an Ackermann mobile robot, wearable drone-control systems, custom PCB development, and autonomous UAV prototypes. Particularly interested in mobile robotics, real-time embedded systems, control systems, and hardware-software integration.

EDUCATION

Bachelor of Applied Engineering, Computer Engineering — 2022 – 2026
Politeknik Elektronika Negeri Surabaya — Surabaya, Indonesia
GPA: 3.73 / 4.00

INTERNSHIP EXPERIENCE

PT. Piranti Kecerdasan Buatan — Aug 2025 – Dec 2025
Research and Development — Tangerang, Indonesia

- Developed an Adaptive Cruise Control system for a 1:10-scale Ackermann mobile robot to maintain a safe following distance and stable speed relative to a preceding vehicle or dynamic obstacle.
- Integrated LiDAR distance measurements and encoder feedback for real-time distance and velocity monitoring.
- Designed and evaluated a Fuzzy Social Force Model-based control strategy with Particle Swarm Optimization to improve following-distance response and speed stability.

PT. Infoglobal Teknologi Semesta — Jan 2025 – Jun 2025
Electronic Development Engineer — Surabaya, Indonesia

- Designed and developed a custom DVR Reader board, including schematic capture and PCB layout using KiCad.
- Developed and tested embedded firmware and supported the migration from ATmega328 to STM32.
- Performed board bring-up, hardware-software integration, functional testing, hardware debugging, and system troubleshooting.
- Prepared technical documentation and validation reports throughout the development process.

TEAM AND ORGANIZATION EXPERIENCE

Cakrawala Skala - PENS UAV Student Research Team — Oct 2024 – Oct 2025
Team Co-Leader and Head of Electrical Division — Surabaya, Indonesia

- Led the Electrical Division in developing UAV and autonomous robotic systems.
- Managed technical planning, task coordination, electrical integration, and system readiness.
- Supervised electrical wiring, hardware assembly, functional testing, and troubleshooting.
- Mentored team members and supported technical decision-making during development and competition preparation.

Cakrawala Skala - PENS UAV Student Research Team — Oct 2023 – Oct 2024
Electrical Division — Surabaya, Indonesia

- Integrated microcontrollers, sensors, actuators, and motor-control components for UAV and autonomous robot development.
- Supported electrical wiring, embedded-system prototyping, hardware debugging, and functional validation.

- Contributed to technical preparation for SAFMC 2024.

Lembaga Minat Bakat PENS Feb 2023 – Feb 2025
Student Resource Development Staff Surabaya, Indonesia

- Supported student-development programs through event planning, documentation, and organizational coordination.
- Served as Chief Organizer of Upgrading 2025, coordinating committee operations and program execution for 80 participants.
- Served as Chief Organizer of Sekolah UKM 2024 and 2025, managing preparation and execution for 24 participants.

AWARDS

2nd Place - Singapore Amazing Flying Machine Competition (SAFMC) 2025 Mar 2025
DSO National Laboratories and Science Centre Singapore, in collaboration with the Ministry of Defence

Semi Finalist - Singapore Amazing Flying Machine Competition (SAFMC) 2024 Mar 2024
DSO National Laboratories and Science Centre Singapore, in collaboration with the Ministry of Defence

Project Funding Grantee Pekan Kreativitas Mahasiswa – Karsa Cipta 2024 Nov 2024
Indonesian Talent Development Center (BPTI), National Center for Excellence (Puspresnas), Ministry of Education, Culture, Research, and Technology of the Republic of Indonesia
Entitled "Hybrid Unmanned Aerial Vehicle (UAV) for Reforestation in Extreme Terrain Using the Seed Bombing Method"

Project Funding Grantee Pekan Kreativitas Mahasiswa – Karsa Cipta 2025 Nov 2025
Indonesian Talent Development Center (BPTI), National Center for Excellence (Puspresnas), Ministry of Education, Culture, Research, and Technology of the Republic of Indonesia
Entitled "Design and Development of an Autonomous Quadcopter Based on Computer Vision for Fire Detection and Extinguishment in High-Rise Buildings"

SKILLS

- **Programming:** C, C++, Python, Embedded C
- **Robotics and Autonomy:** ROS, ROS 2, Gazebo, SLAM, Navigation, LiDAR Integration, Ackermann Kinematics
- **Embedded Systems:** STM32, ESP32, ATmega328, FreeRTOS, STM32CubeIDE, Arduino IDE
- **Electronics and PCB Design:** KiCad, Schematic Capture, PCB Design, Power Distribution, Electrical Wiring, Hardware Bring-up, Soldering, Oscilloscope
- **Communication and Interfaces:** UART, SPI, I²C, PWM, Wi-Fi UDP, CRSF, Sensor–Actuator Interfacing
- **Control Systems:** Fuzzy Logic, Social Force Model, Particle Swarm Optimization, Adaptive Cruise Control, Control-System Integration
- **Development Tools:** Git, Linux, Docker